

Embed an ANN-driven adaptive droop controller in the rooftop solar inverter that continuously modifies droop coefficients based on ambient temperature and load conditions to limit overcurrent and heat generation

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'thermal degradation' concerning rooftop solar inverter. User-submitted engineering problem (session input): "Why do commercial rooftop solar inverters fail prematurely in hot climates?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Physics-of-Failure (PoF) based reliability assessment for photovoltaic (PV) inverters requires long-duration electrical and electrothermal stress histories, yet generating such stress histories with h'; 'Growth in power requirements and the complexity of emerging distribution systems are creating the need for new solid-state products for many applications. The qualification requirements, testing stand'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Adaptive droop characteristic adjustment via ANN-assisted real-time tuning of droop coefficients and PI gains

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Embed an ANN-driven adaptive droop controller in the rooftop solar inverter that continuously modifies droop coefficients based on ambient temperature and load conditions to limit overcurrent and heat generation

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: h, dT.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected). Engaged variables: k, dT/dx.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.